

Game Design Document – Snake Escape

1. Game Overview

One-Sentence Pitch

Tap snakes on a grid and guide them off the board — the satisfying puzzle of clearing a tangled snake pile without making a mistake.

Game Concept

Snake Escape is a mobile casual-puzzle game. Players are presented with a dot-grid board filled with colourful snakes facing different directions. Tapping a snake slides it off-screen along its exit path — if the path is clear. When blocked by another snake or an obstacle, the move costs health. The goal: clear all snakes from the board with at least 1 health remaining.

Platform & Target Audience

Field	Value
Platform	iOS / Android (mobile)
Target Audience	Casual puzzle players, ages 16–45
Session Length	2–5 minutes per level
Core Audience Fantasy	“I am smart — I can solve this.”

Game Pillars

Pillar	Description
Satisfying Clarity	Every move has an obvious, clean result — no ambiguous outcomes.
Escalating Complexity	Difficulty scales smoothly: small boards → larger boards, more snakes, obstacles.
Low Frustration Ceiling	Failure is forgiving: boosters, revives, and restarts keep the game feel positive.
Daily Habit	Multiple daily engagement loops ensure players return every 24 hours.

2. Core Gameplay Mechanics

2.1 Board System

Grid type: Dot-grid. Dots define snake positions and bend paths.

Board size range: Minimum 3×3, maximum 40×40.

Players can pinch-to-zoom on larger boards (two-finger gesture).

2.2 Snakes

A snake occupies a sequence of dots. 4 segments = 2 dots.

Each snake has a facing direction: Left / Right / Up / Down.

Direction determines the exit trajectory when tapped.

Snakes render with curves/bends along their dot paths.

2.3 Tap-to-Move

Correct move: Snake’s path to the board edge is unobstructed → snake slides off-screen with exit animation → removed from board.

Wrong move: Snake’s path is blocked by another snake or obstacle → snake cannot exit → -1 health.

Rule: only the first wrong tap on a blocked snake costs health (repeated taps on same blocked snake do not re-penalise).

2.4 Health / Lives System

Each level has configurable health (default: 3).

Health = number of wrong moves allowed before level failure.

Health = 0 → level fails → Revive or Failed screen is shown.

2.5 Level Completion & Failure

State	Trigger	Result
Win	All snakes exited	Show Result screen, award Coin
Fail	Health = 0	Show Revive/Failed popup
Revive	Player pays 300 Coin or watches Ad	Restore health, continue level
Restart	Player chooses Restart	Reset level; lose progress

3. Obstacles & Support Elements

3.1 Obstacle: Wall (Countdown)

A Wall blocks any snake that tries to cross it.
Each snake that successfully escapes the board reduces the Wall countdown by 1.
When countdown reaches 0, the Wall disappears.
Collision with a Wall costs health and triggers a collision animation.

3.2 Support Element: Black Hole

A Black Hole acts as a portal: a snake entering one Black Hole exits from the paired Black Hole (or directly off the board).
Teleport animation: swirl/shrink on enter, expand on exit.

3.3 Extensibility

The element system is designed for future additions: moving barriers, ice tiles, locked gates, etc.
First-encounter popup for any element: icon, description, “OK” button with pulse animation.

4. Booster System

4.1 Hint

Field	Value
Effect	Highlights one snake that can currently escape
Cost	300 Coin or watch 1 Ad
Limit	Per use (inventory-based)

4.2 Grid Vision (Ruler)

Field	Value
Effect	Shows directional exit arrows on all snakes simultaneously
Cost	900 Coin or watch 1 Ad
Limit	Per use (inventory-based)

4.3 Booster Acquisition Sources

IAP purchases (Shop)
Daily Missions rewards
Daily Spin rewards
Tournament rewards
No Ads bundle

4.4 Booster UX Flow — First Introduction

Popup: “New Booster” (icon + name + description)
Player taps OK → popup closes
Dark overlay appears → animated finger points at the booster button

Player taps booster → effect activates → inventory −1

5. Tutorial & Onboarding

Progressive tutorial triggers at specific level milestones or first-encounter events. Each tutorial fades out after player completes the prompted action — never blocking gameplay.

Tutorial	Trigger	Content
1. Tap to Move	Level 1	Animated finger over a snake. Text: “Tap to move!” Fade on first correct tap.
2. Pinch to Zoom	First large board	Two-finger pinch animation. Text: “Use two fingers to zoom in and out.”
3. Hint Booster	First Hint available	Booster intro popup → OK → dark overlay → finger at Hint button.
4. Grid Vision Booster	First Ruler available	Same flow as Tutorial 3 for Grid Vision.
5. Wall Obstacle	First Wall level	Popup: Wall illustration + description “Don’t hit the wall!”
6. Black Hole Element	First Black Hole level	Popup: Black Hole illustration + description.

Finger Animation Spec

State	Duration
Default (idle)	Loop
Raise	400ms (hold 33ms)
Tap down	200ms (hold 33ms)
Return to default	167ms

6. Progression System

6.1 Level Structure

Levels are numbered sequentially. No chapter wall between levels.
Difficulty curve: board size grows, snake count increases, obstacles introduced, fewer health margins.
Each level belongs to a Chapter (thematic grouping) for visual variety.

6.2 Level Unlock

Levels unlock sequentially upon completion.
Tournament joins at Level 14 — first competitive gate.

6.3 Difficulty Dimensions

Dimension	Easy	Medium	Hard
Board size	3×3 – 6×6	7×7 – 12×12	13×13+
Snake count	2–4	5–8	9+
Obstacles	None	1 Wall	2+ Walls / Black Holes
Health	3+	3	2–3

7. Economy & Monetization

7.1 Currency

Single currency: Coin. No premium currency tier.

7.2 Coin Sources (Earn)

Source	Amount
Level completion	40 Coin
Level completion + Ad	80 Coin (×2 offer on Result screen)
Daily Video Bar: 1 ad	100 Coin
Daily Video Bar: 3 ads	300 Coin + 1× Grid Vision
Daily Video Bar: 5 ads	500 Coin + 1× Hint
Daily Mission: Complete levels	50–500 Coin
Daily Mission: Collect snakes	50–500 Coin
Daily Spin	300–1,200 Coin or 1× booster
IAP Coin Packs	1,000–100,000 Coin

7.3 Coin Sinks (Spend)

Spend	Cost
Hint booster	300 Coin
Grid Vision booster	900 Coin
Revive	300 Coin

7.4 IAP Products

Product	Contents	Price
Coin Pack — Starter	1,000 Coin	\$1.99
Coin Pack — Small	5,000 Coin	\$7.99
Coin Pack — Medium	10,000 Coin	\$14.99
Coin Pack — Large	25,000 Coin	\$29.99
Coin Pack — XL	50,000 Coin	\$49.99
Coin Pack — Mega	100,000 Coin	\$79.99
No Ads Bundle	Remove ads + 3,000 Coin + 3× Ruler + 3× Hint	\$12.99
No Ads Only	Remove ads	\$7.99 (50% promo)
Seasonal Bundle (e.g. Santa's Giftbox)	3,000 Coin + 3× Ruler + 3× Hint	\$8.99 (30% off)

7.5 Ads

Rewarded: Coin doubler on result, Revive, booster fallback, extra Daily Spins, Daily Video Bar.

Interstitial: Between levels (frequency capped; disabled if No Ads purchased).

Banner: On dashboard/menus (disabled if No Ads purchased).

8. Daily Engagement Loops

8.1 Daily Mission System

7 mission types assigned daily; reset each 24h (server-side).

Missions displayed on Dashboard (highest-priority incomplete one shown as preview).

Mission	Tier	Reward
Complete (x) Levels	1/3/5/7/10	50–500 Coin
Collect (x) Snakes	50/100/200/300/500	50–500 Coin
Use (x) Hints	1/2/3	300–900 Coin
Use (x) Grid Vision	1/2/3	100–300 Coin
Spend (x) Coin	300/600/900	Grid Vision or Hint booster
Collect (x) Coin	300/600/900	Grid Vision or Hint booster

8.2 Daily Spin (Lucky Wheel)

1 free spin per day (server-side reset).

Extra spins: watch Ad (max 5 spins/day total).

6 segments, equal probability (16.67% each): 300 / 600 / 900 / 1,200 Coin / 1× Grid Vision / 1× Hint.

Spin animation: deceleration → segment highlight → prize popup → inventory update.

8.3 Daily Challenge

One unique curated puzzle served by backend daily for all players.

Calendar UI: shows today's challenge, completed days, and replayable past days.

Trophy Room: Monthly trophy awarded for completing all 30/31 challenges.

Result variants: “Completed” / “Well Done” with reward.

Replay missed days (possibly via Ad watch).

8.4 Daily Reward (Login Bonus)

Consecutive login rewards with chest opening animation.

Returning player “Welcome Back” screen with bonus.

8.5 Daily Video Bar

Progressive ad bar: watch 1 / 3 / 5 ads per day for increasing rewards (see §7.2).

9. Competitive Features

9.1 Tournament System

Join trigger: Level 14 completion popup.

Scoring: Win a level → Score += BaseScore(1) × StreakMultiplier. Multiplier increments after score is added.

Win Streak Multiplier Table:

Streak	Multiplier
1	×1
2	×2
3	×4
4	×6
5+	×10

Streak reset: On level reset, restart after game over, or reopening with reset. Preserved when starting from Dashboard or Win screen.

Leaderboard: Rankings, scores, streak bar. Updated server-side.

Reward tiers:

Rank	Reward
1	9,000 Coin + 9× Hint + 9× Grid Vision
2	6,000 Coin + 6× Hint + 6× Grid Vision
3	3,000 Coin + 3× Hint + 3× Grid Vision
4–10	2,000 Coin + 2× Hint + 2× Grid Vision
11–100	1,000 Coin + 1× Hint + 1× Grid Vision
101+	500 Coin + 1× Hint + 1× Grid Vision

9.2 Play With Friend

Asynchronous (or real-time) match: both players complete the same level.

Scoring: TotalScore = BaseScore(100) + TimeBonus (TimeBonus = remaining countdown seconds).

Countdown: 3 minutes visible on gameplay HUD.

Result screen: winner declared, score breakdown, snake count comparison.

Requires friend list management (add/remove) and invitation flow.

10. UI/UX Specification

10.1 Screen Inventory

Screen	Key Elements
Splash / Loading	Logo, loading bar
Home / Dashboard	Play button, current level #, Coin balance, booster quantities, feature entry points (Tournament, Daily Spin, Daily Challenge, Daily Mission preview)
Gameplay HUD	Health hearts, Coin counter, Hint + Grid Vision buttons (with quantity), level #, pause button, optional timer
Result Screen (V1 & V2)	Coins earned, ×2 ad offer, Next Level button — two visual variants
Revive / Failed (V1 & V2)	Revive (300 Coin / Ad), Restart, Home — heart-mode and time-mode variants
Tutorial Overlays	Animated finger, text boxes, dark overlay, popup containers
Shop	Tabs: Coin Packs, Bundles, Promotions
Settings / Pause / Language	Sound, music, vibration toggles; language selector; Facebook link
Daily Mission Screen	Quest list with progress bars and Claim buttons
Daily Spin Screen	Wheel, spin button, ad-for-spin button
Daily Challenge Screen	Monthly calendar, Trophy Room, result screens
Daily Reward Screen	Chest open animation, consecutive day tracking
Tournament Screens	Join popup, leaderboard, streak bar, reward screen, tutorial popup
Play With Friend Screens	Friend invite, match setup, in-game overlay, score comparison
Friend List	Search, add/remove, online status
Booster Intro/Buy Popup	New booster popup + in-level purchase prompt
No Ads Screen	Offer detail with pricing
Rating Prompt	App store review request
Welcome Back	Return bonus notification

10.2 Navigation Flow (High Level)

Splash → Home (Dashboard)

|-- Tap Play → Gameplay → Result / Revive / Failed

|-- Shop

|-- Settings / Pause

|-- Daily Mission

|-- Daily Spin

|-- Daily Challenge

|-- Tournament (unlocks at Level 14)

|-- Play With Friend

10.3 Localisation & Themes

Localisation: Multi-language support. Portuguese confirmed in assets. Design for text expansion (up to 30% longer strings).

Dark Mode: Full dark-mode visual variant exists in art assets (UI_Mobile_Darkmode).

Landscape Layout: Landscape orientation variant available (Ui_Landscape).

11. Animation Specifications

Animation	Spec
Snake movement speed	1 dot = 33ms
Snake exit	Slide off-screen with acceleration + particle trail
Wrong move feedback	Bounce back + screen shake + heart break animation
Tutorial finger	4-state loop (see §5)
Popup	Fade In / Fade Out. OK button pulse animation.
Booster: Hint	Glow effect on the hinted snake
Booster: Grid Vision	Arrow reveal animation across all snakes
Wall countdown	Number decrement feedback. Wall shatter FX at 0.
Black Hole teleport	Swirl/shrink enter → expand exit at destination
Spin wheel	Smooth deceleration → segment highlight → prize popup
Win streak	Visual multiplier climb 1→10× with celebratory particles

12. Backend & Infrastructure Requirements

12.1 Player Data (Cloud Save)

Persist: level progress, Coin balance, booster inventory, settings, mission progress, tournament score.

Sync on session start; local cache to protect 60fps gameplay.

12.2 Daily Reset Service

Server-side UTC reset for: Daily Missions, Daily Spin count, Daily Video Bar count, Daily Challenge.

12.3 Tournament Lifecycle

Server manages: start/end timestamps, score submission, real-time leaderboard, reward distribution.

12.4 Daily Challenge Service

Backend serves one unique level JSON per day to all players.

Supports replay requests for past days.

12.5 Friend System

Friend requests, storage, invitation flow, real-time or async score sync for Play With Friend.

12.6 Analytics Events (Required Tracking)

Event	When
level_start	Player starts a level
level_complete	Player wins a level
level_fail	Health reaches 0
booster_used	Player uses Hint or Grid Vision
revive_used	Player revives
iap_purchased	IAP transaction completes
ad_watched	Rewarded ad completes
session_start	App opens

12.7 Remote Config / Live Ops

Economy values (prices, rewards, probabilities) configurable remotely.

Tournament schedules and reward tables via remote config.

Feature flags for seasonal events (Seasonal Bundle).

12.8 Push Notifications

Notification	Trigger
Tournament ending	1 hour before tournament closes
Daily Challenge reminder	Daily, if not completed
Welcome Back	After 2+ days absent
Streak at risk	If tournament streak active but no play today

13. Technical Requirements

See `project-context.md` for binding constraints. Summary below.

Requirement	Constraint
Engine	Unity 6, URP
DI	VContainer — all services injected via <code>[Inject] Construct()</code> . No <code>FindObjectOfType</code> .
Reactive State	UniRx <code>ReactiveProperty<T></code> for all mutable state. UI binds via subscriptions.
Async	UniTask — no <code>IEnumerator</code> , no <code>System.Threading.Tasks.Task</code> .
Performance	Core gameplay loop must produce zero GC allocations per frame.

Requirement	Constraint
Main Thread	Level solvability validation and cloud API calls must run on background threads via <code>UniTask.Run()</code> .
Memory	Dispose all subscriptions via <code>CompositeDisposable</code> in <code>OnDestroy()</code> .
Input	Unity New Input System (configured). No <code>Input.GetKey</code> .
Framework services	Audio, Ads, Analytics, Economy, Notifications via <code>sun.*</code> package interfaces.

14. Level Design Pipeline

14.1 Level Data Format

Serialised level file contains: board size, dot positions, snake placements (start dot + direction), obstacle placements, health override.

14.2 Level Editor

Unity Editor tool for designers: place dots, position snakes, set directions, add obstacles, configure health per level.

14.3 Level Validation

Automated solvability checker runs on background thread (`UniTask`): verifies at least one valid sequence of moves exists to clear all snakes. Must pass before level is shipped.

14.4 Chapter Organisation

Levels grouped into themed chapters. Chapter boundaries are cosmetic (visual transition) — no content lock.

15. Scope Summary

Metric	Count
Total Epics	14
Total User Stories	~106
UI Screens	26+ folders / variants
IAP Products	9 (6 Coin packs + 3 bundles)
Daily Engagement Loops	5 (Mission, Spin, Challenge, Reward, Video Bar)
Competitive Features	2 (Tournament, Play With Friend)
Obstacle Types (v1)	2 (Wall, Black Hole)
Booster Types (v1)	2 (Hint, Grid Vision)

IMPORTANT: Daily Challenge (Epic 10) has extensive UI assets (Calendar, Trophy Room, Results) but minimal GDD text. Formal specs align to the existing art direction documented here.

NOTE: Daily Reward (login bonus / chest animation) exists in art assets but had no prior GDD section. It is now captured in §8.4 and Epic 12.

TIP: Dark Mode and Landscape variants are present in art assets. These should be scoped as explicit Epic 12 stories to ensure implementation coverage.